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1 Introduction

Travel is becoming easier and easier for the common folk. The newest trend in personal travel are in the form of electric scooters, amidst bike and car sharing services.

Scooters are also gaining popularity within major cities, following their introduction in major cities such as Austin, San Francisco, and Santa Monica. They are a great choice for completing the last mile of a consumers' commute; however, the booking process is not as easy.

The issue that has arisen within the scooters' application is that there are 10 or more different companies, which leads to 10 or more different apps being required to be downloaded on a user's mobile device. Furthermore, one might find that a scooter that they usually use from a particular company might be out of their reach. Therefore this necessitated the search through 9 other applications to find the scooter that was available, charged, and close at the same time.

After exploring various options, there was finally an app released on the market that had put all these scooters in one place: Transit. This application was great, until it wasn't, as it did not allow the user to rent the scooter from their app. Instead, users found themselves being redirected to the scooter company of choice's app page to download the app to complete the transaction, which brings the issue faced back to square one.

This is a major inconvenience since every customer wants the easiest access to something, and two button clicks is always worse than one. Not to mention that the scooter that they originally wanted may no longer be available given the extra time wasted having to download an additional application, registration, ID verification, and payment insertion.

2 Solution

As a result of the pain point discussed in the introduction, the solution proposed to solve this problem is *ScooterDragon*, or one might think of it as, the kayak for scooters.

This is an application that will incorporate all the major companies apps into one, single, user-friendly app, which will allow users to rent the scooter straight from the application itself without having the need for 10 different apps from 10 different companies.

This permits customers to have less apps on their phones, and less button clicks to be able to find the ideal scooter. Furthermore, the integrated GPS system will locate not only the closest scooter, but also the most economical one within the given area.

Specific user settings will be implemented. Should the user only care about the ease of access, they will be able choose this setting to sort the scooters by price, or location.

No matter the price of the scooter, possible rental options of the closest will scooter will be shown. On the flip side, if a consumer prefers to be economical, an option to search for the cheapest scooter in a certain pre-set radius is implemented. This app allows consumers to have the best of both worlds, albeit with a small overhead fee applied.

Details of *ScooterDragon* will be discussed in the following sections.

3 Target Market

3.1 Scooter Companies

As ScooterDragon does not have any scooters of its own, the partnership between us and other scooter companies is paramount. Companies such as Lime and Bird are two major scooter companies available in the United States.

Other companies, such as Uber and Lyft have also stuck their hand in the scooter game. All these companies are targets for ScooterDragon. The more companies that agree to partner, the more accessibility the consumer has.

ScooterDragon will also allow smaller scooter companies to be more competitive, and possibly get some new companies into the game. This would also give these companies a competitive playing field, given that the lower their prices are, the further up top they would appear in the search ranking.

3.2 End-User Demographics

To better understand the users of *ScooterDragon*, a survey was conducted for a month to study the current behavior and characteristics of a user's scooter reservation process. The motivation behind the choice of a certain company over the other, as well as the issues and problems faced were asked.

This was important as an application designed without the end-user in mind will impede the growth and sustainability of the business. It was crucial to understand the users' behaviors, and leverage on the issues that are currently being faced to be able to address them in the application for the market.

Methods

A form was created online, to which the link was forwarded to all possible contacts within the network of the authors. The link was also shared within the UIC community for students to participate in. Some key questions asked included:

1. User age
2. Favored scooter companies
3. Problems encountered when completing the transaction
4. What could have been done to improve their reservation experience
5. Willingness to pay the overhead costs based on the business model proposed if it guarantees convenience

Results

A form was created online, to which the link was forwarded to all possible contacts within the network of the authors. The link was also shared within the UIC community for students to participate in. Some key questions asked included:

1. Reflect and scale accordingly should a larger sample size have been obtained.
2. Be similar in other cities with different permutations of scooter companies available

The results gathered showed that:

1. Majority of users (86%) ranged from 21-25 years old. No data was gathered for users below 18 since state laws require scooter users to be a minimum of 18 years.
2. When asked about their favorite or 'go to' scooter company, the results gathered were scattered. Even though Lyft showed a higher percentage of the results, this was not representative of its actual market share.

Furthermore, the scattered nature of the company choices was an especially promising set of data since this is a reflection of what ScooterDragon hopes to exactly resolve –

to bring all these different applications from various companies into a single integrated system, so as to eliminate the need for ten applications when ScooterDragon requires only one without compromising on selection for the end user.

3. The general consensus given by participants for choosing a certain app included the company's application being the first on the app store, as well as ease of use. Many also agreed that convenience and 'sheer laziness' was the sole reason they had the application.

This evidence suggests that when designing the interface of ScooterDragon, user experience will be held paramount; However, furnishing these details in depth is out of scope of this paper.

4. 93% of users supported the idea of paying the 10¢ and 1¢ overhead fees for unlocking and per minute usage of the scooters respectively. The results are coherent with the results obtained in (3) such that convenience and ease of use are highly desired by the end-users.

This also proves that the market, in pursuit of convenience, is willing to pay a small additional fee on top of the baseline prices set by the respective scooter companies.

Raw data collected from the survey is available in Appendix A.

4 Competition & Strategic Partners

4.1 TransitApp

As the only mass scooter app on the market now in the United States, Transit yields to be the biggest competitor for ScooterDragon. However, Transit does not allow the same flexibility as ScooterDragon will. Transit does not allow users to rent the scooter directly from the app, as it redirects the consumer into the companies own app. Transit also struggles to have good GPS location of the scooters. Many users have complained that there will be a scooter shown on the map, but when the customer arrived it was no longer there.

Partnering with Transit could be useful as they have the baseline coding laid out for the system, and since ScooterDragon will be partnering with the scooter companies directly as well, the GPS systems will be more precise as they are coming directly from the company.

4.2 Jelbi

While the proposed idea was under conception, Jelbi was launched on September 24, 2019 (less than two months ago). Jelbi was Berlin's answer in the form of ScooterDragon. The application integrates not just scooters, but all forms of transport including buses, trains, ride-sharing services, and even ferry passes.

Within two months, the application has had over 10,000 downloads^[3] in the Google Play App Store alone. Data was not released by the Apple app store with regards to the number since that was information only made available to the app developers themselves.^[4]

While the app has received much fanfare and a high volume of downloads, there were a multitude of problems that users faced. Apart from the technical issues, the major problem lied in the smooth transitioning of services delivered, as many users were unable to buy ferry passes despite Jelbi promoting that it could. Another issue that affected mostly tourists were the lack of payment options. Currently, only Visa and Mastercard is an accepted form

of payment. As such, users having American Express cards found that they were heavily restricted.

Despite the fact that Jelbi is offering the same intended services as ScooterDragon, it is not a cause of concern at the moment because Jelbi is headquartered in Berlin, Germany. It would only make sense for them to introduce the application to other major cities within Germany, such as Frankfurt, Munich, Dusseldorf, etc. before expanding out to other countries in the EU which is not even under trial or consideration at the moment. Hence, Jelbi will primarily seek to serve the EU market, while ScooterDragon will primarily seek on serving the US market.

5 Business Model

As mentioned in the solution, ScooterDragon seeks to adopt the model that KAYAK.com currently has; that is to be, the search engine for all scooters within the city limits which shows users scooters that are nearest to them, sorted according to price and battery level.

5.1 KAYAK Model

KAYAK was launched in 2004 by co-founder Steve Hafner^[1], who initially founded Ortiz in response to his dislikes for travel agencies. However, Hafner failed to realize that Ortiz itself was just another form of a travel agent, albeit an electronic one. Not only did Ortiz defeat the entire purpose of his intention, research also showed that successful transactions only take place between 2-6% of the time when visitors utilized the website to fulfil their travel booking transactions.^[2]

Hafner realized that he had deviated away from the original mission, which was not to carry any function of a travel agent when visitors needed to book tickets for their flights or hotels. His response to that? KAYAK. The website was designed to only focus on providing leads for various airline and ticket options without having to physically handle the transaction of selling those tickets.

As such, Hafner employed the following business model for KAYAK^[1]:

1. To function as the ‘Google’ search engine for traveling related searches. When a customer is ready to buy a ticket, the website then redirects the customer to the appropriate website to complete the transaction.

By omitting this step, KAYAK was therefore able to operate more efficiently, especially in terms of financial costs since they did not need handling agents or customer service representatives for their services.

2. To be a free-to-use platform for their users. To ensure that the company focused on

the user centric experience, KAYAK eliminated the need for transaction fees.

3. Distribution revenue as a first source of income from various travel and airline websites. For example, when a customer completes a transaction on Delta Airlines (DAL) as a result of KAYAK's redirection, DAL then pays KAYAK a portion of the ticket's price as commission. Collectively, airline commissions contribute 30% of KAYAK's revenue source.
4. Advertisements as a second source of income. KAYAK offers advertisements to various travel sites, experience packages, and other related businesses and obtains its income via a cost per click or cost-per-impression rate, which are similar to various strategies adopted by many websites with advertisements.

It is inarguably evident that KAYAK's ability to focus on delivering the experience to its customers while doing away with middleman costs contributed to its success. Kayak was bought out by Priceline in 2012 for 1.8 billion dollars^[5], but before that Kayak a net worth of \$850 million dollars.^[6]

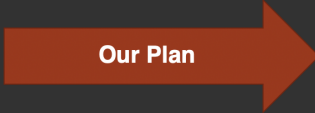
Furthermore, given its size as an online travel search platform, it has enticed many airlines and hotels to be part of the search as these companies were then able to offer a more competitive pricing. The lower the price for a ticket or hotel stays, it meant that their company's products would be offered at the top of the list finishing a search by the end user.

5.2 ScooterDragon

The only difference with ScooterDragon is that, unlike KAYAK, the application will handle the booking directly within the app itself, instead of redirecting users to the various scooter companies like Transit is already implementing.

Here is how ScooterDragon is going to work:

Table 1: Price comparison before and after implementation of the overhead cost

Cost			Cost	
Bird	\$1 to Unlock		Bird	\$1.10 to Unlock
	\$0.15 - \$0.39/minute			\$0.16 - \$0.40/minute
Lime	\$1 to Unlock		Lime	\$1.10 to Unlock
	\$0.15 - \$0.31/minute			\$0.16 - \$0.32/minute
Uber Scooter	\$1 to Unlock		Uber Scooter	\$1.10 to Unlock
	\$0.15 - \$0.30/minute			\$0.16 - \$0.31/minute
Lyft Scooter	\$1 to Unlock		Lyft Scooter	\$1.10 to Unlock
	\$0.15 - \$0.29/minute			\$0.16 - \$0.30/minute

As shown in the table (1) above, there is a slight increase in price with the overhead costs. Riders will pay an additional 10¢ overhead unlocking fee, as well as an additional 1¢ per minute riding fee. Note that the total cost of each trip may differ slightly, as it is ultimately dependent on the scooter company's rental charges.

This additional amount that consumers pay for convenience is a reasonable amount, as previously stated by 93% of users who were in favor; thus, showing that the time saved is worth the extra penny.

6 Future Developments

The future is bright, especially for a world and generation that relies heavily on app usage. To date, there are more than 2.5 million apps available for download on the Google Play store alone.^[6] This suggests that applications are built to serve the users in their daily lives, and ScooterDragon also exists to do the same.

Since the only variable that needs updating and building lies with coding, ScooterDragon will thus be able to adapt to the changing trends of the seasons as well. At present, there are three proposed upgrades for consideration. They include:

1. Food Pick-up Integration

An issue that is apparent with scooters now is that to pick up food a customer has only two options. The customer has to leave the scooter unattended for an extended period while the food is being prepared, which is dangerous to both the user, and the company, as an unattended unlocked scooter can be stolen, and will work until the scooter runs out of battery.

This will cost the customer money as the stolen scooter's time is still under their account. The other option customers have is to lock the scooter, but the main issue with this option is risking the loss of the scooter to someone else who reserved it. But, there is a less thought about issue with regards to the the unlocking fee.

As previously stated right now every time a scooter is unlocked it costs the customer a flat unlock fee. If a customer has a short trip home, but they want to stop for food, this will almost double the cost. Allowing the customer to order ahead through the app, specifically direct the route by the restaurant, and have the food ready by the time the customer gets there.

2. Grocery Pick-Up Integration

The next idea would be grocery store pick-up. This is integrating the previous idea that is already in place by stores such as Target, or Walmart. The initial roadblock for this issue would be the limited capacity for scooters to hold groceries.

Without the integration of a basket on scooter, as divvy has done on bikes, this idea is slightly farfetched. However, even without the basket, two bags would be able to be held on one scooter. This would create an easy and fast way for people without a car to get their groceries.

3. More Payment Options

The final idea for the moment, and the most necessary is the integration of more payment methods. It was apparent in Jelbi where there were many complaints about people not being able to get their cards working to rent out the scooters. Credit cards also imply fees on the companies. If ScooterDragon would be able to include more options such as PayPal or Venmo this would allow more users to be able to rent and do it faster.

With scooters growing in popularity, and peoples need for speed always increasing, ScooterDragon aims to be able to facilitate both. ScooterDragon will allow customers a fast and efficient way to rent, while eliminating the need for 10 different applications for each scooter company. With the scooter companies on board, ScooterDragon will be the number one stop for all your scooter needs and perhaps, even more.

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8 Appendix A: Survey Data

Figure 1: User Age Demographics

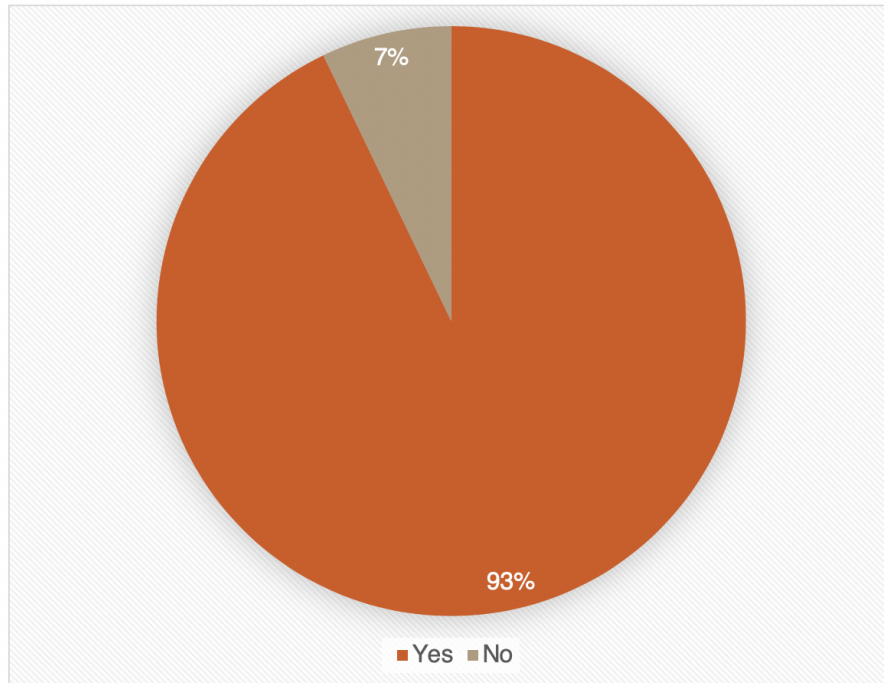


Figure 2: Preferred Scooter Companies

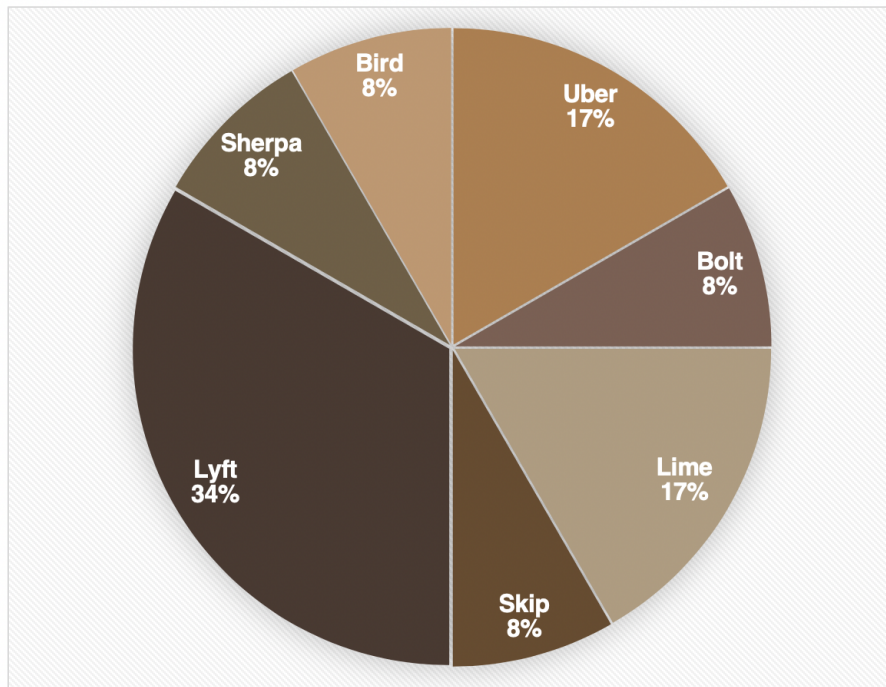


Figure 3: Willingness to pay overhead fee

